NLP Lecture Notes #1.3: Dictionaries for Text Analysis

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In computational text analysis, a **dictionary**, also known as a **lexicon**, is one of the simplest yet most powerful tools. It is not a dictionary in the traditional sense of providing definitions, but rather a curated list of words associated with a specific category or value.

Definition

An NLP **dictionary** or **lexicon** is a list of words (or terms) that are used to measure a particular concept in a text. Each word in the list evokes a certain category, and may optionally be paired with a score or rating on a scale.

This method is popular because it is "easy and cheap to apply to a variety of problems". The core idea is to count the occurrences of words from the dictionary within a document to get a score for the concept the dictionary represents.

Example

A Simple Sentiment Dictionary We could create a basic dictionary to measure positive and negative sentiment in a movie review.

- **Positive Words**: {'great', 'amazing', 'love', 'excellent', 'happy'}
- **Negative Words**: {'bad', 'terrible', 'hate', 'awful', 'boring'}

To analyze the review "This movie was amazing, not boring at all!", we would count:

- Positive score: 1 ('amazing')
- Negative score: 1 ('boring')

This simple count-based approach can provide a quick measure of the overall tone of the text. (Note: Handling negation like "not boring" is a more advanced challenge for this method).

While sentiment analysis is the most common application, dictionaries can be used to measure any categorical concept.

- **Sentiment and Emotion**: Classifying text as positive, negative, or neutral, or identifying specific emotions like joy, anger, or sadness (e.g., the NRC Word-Emotion Association Lexicon).
- **Economic Confidence**: Tracking public opinion by counting words related to the economy, such as {'economy', 'job', 'jobs'}, in social media posts like tweets.
- **Psychological States**: The LIWC (Linguistic Inquiry and Word Count) dictionary contains lexicons for concepts like inhibition, insight, and family dynamics, often used in social psychology.
- **Connotation**: Analyzing the implicit positive or negative attitudes conveyed by verbs. For example, the verb "suffer" connotes a negative experience for its subject (agent).

Human Labeling (Hand-Crafted) This is the most direct method, where humans annotate words based on the dimension of interest.

- **Process**: For each word in a candidate list, human annotators (e.g., on crowdsourcing platforms) are asked to provide a judgment.
- **Scales**: Judgments can be categorical (e.g., positive vs. negative) or on a scale. *Likert Scales*: Annotators rate association, e.g., {not, weakly, moderately, strongly} associated with "joy". *Numerical Scales*: Annotators provide a score, such as -5 (most negative) to +5 (most positive), like in the AFINN lexicon.

Semi-Supervised Expansion This method starts with a small set of "seed" words and automatically expands the dictionary to include similar words.

1. **Seed Set**: Define a small, high-confidence set of words representing the concept. For sentiment, this could be {'good', 'excellent'} for positive and {'bad', 'terrible'} for negative.
2. **Similarity Metric**: Use a word representation model (like word embeddings, which we will cover later) to find words that are semantically similar to the words already in the lexicon.
3. **Expansion**: Add the most similar words to the dictionary and repeat the process. For example, if 'awesome' is found to be very similar to 'good', it is added to the positive list.

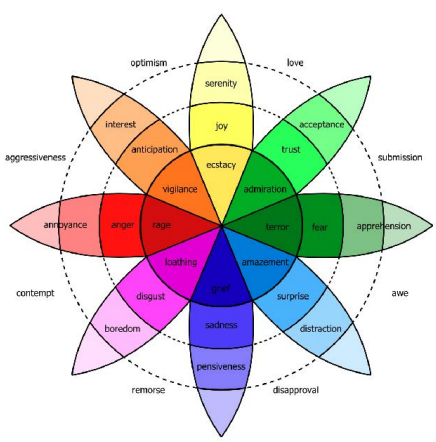

Figure 1: a scatter plot with emotional labels ("love," "like," "awesome," "fleek," "hate," "terrible") plotted on a coordinate system, possibly representing a semi-supervised expansion of emotional categories based on two axes (e.g., positive/negative and intensity).

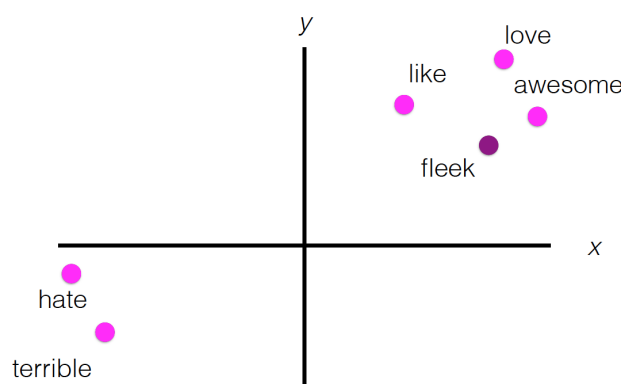

Figure 2: a model that categorizes human emotions into a spectrum, with primary emotions like joy, sadness, anger, and fear at the center, branching out into more nuanced feelings (e.g., ecstasy, grief, rage, awe) as you move outward.

Supervised Learning This method uses a labeled dataset (e.g., documents with ratings) to learn which words are most associated with a particular label.

- **Process**: Given a corpus of texts with labels (e.g., movie reviews with 1-5 star ratings), we can calculate the statistical association of each word with each label.
- **Calculation**: A simple measure is the probability of a word given a class, normalized across all classes. For a word w and a class c:

$$\text{Score}(w, c) = \frac{P(w|c)}{\sum_{c'} P(w|c')}$$

A word like "excellent" will have a high probability (and thus a high score) for the 5-star class, while "disappointing" will have a high probability for the 1-star class.

Caveats and Limitations While powerful, dictionary-based methods have important limitations.

- **Type vs. Token Distinction**: Dictionaries operate on word **types**, not **tokens**. This means they ignore the context in which a word is used. For example, the word "sick" could mean "ill" (negative) or "awesome" (positive), but a dictionary will assign it a single, fixed score.
- **Domain Dependence**: The sentiment of a word can change drastically between domains.

  Example

  The word **"crude"** is generally negative in everyday language. However, in a financial earnings report for an oil company, "crude" is a neutral, descriptive term. Applying a general sentiment dictionary to a financial corpus can lead to incorrect results.

- **Negation and Modifiers**: Simple dictionary methods struggle with linguistic phenomena like negation ("not good"), intensifiers ("very good"), and attenuators ("somewhat good").

Despite these limitations, dictionaries remain a fundamental tool for their simplicity, interpretability, and effectiveness in many text analysis scenarios.

Example

Dictionary Application You are given the following mini-dictionary for emotions:

- **Joy**: {'happy', 'joyful', 'delighted', 'celebrate'}
- **Sadness**: {'sad', 'cry', 'grief', 'unhappy'}

Analyze the following sentence and calculate its Joy and Sadness scores based on a simple word count: *"I was so happy to celebrate with my family, but I still cry when I think of our loss. It's a sad kind of grief."*

The sentence is: *"I was so happy to celebrate with my family, but I still cry when I think of our loss. It's a sad kind of grief."*

- **Joy Words Found**: 'happy', 'celebrate'.
- **Sadness Words Found**: 'cry', 'sad', 'grief'.

Scores:

- **Joy Score**: 2
- **Sadness Score**: 3

Based on this simple dictionary method, the sentence has a higher Sadness score than a Joy score, reflecting its mixed emotional content.

Example

Domain Dependence Imagine you are building a dictionary to measure "excitement" in the context of video game reviews. You start with a general seed set: {'exciting', 'thrilling'}. A semi-supervised method suggests adding the word 'grind'. Why might this be a poor addition to your dictionary?

The problem is **domain dependence**.

- In general English, "grind" can have negative connotations (e.g., "the daily grind," meaning monotonous work).

- In the specific domain of video games, "grind" or "grinding" refers to the act of repeating a task over and over to gain experience points or items. This is almost universally seen as a **boring, tedious, and unexciting** part of a game.

Therefore, adding 'grind' to a dictionary for "excitement" would be incorrect. It is semantically related to the topic of gaming, which is why a semi-supervised method might find it, but its connotation within that domain is the opposite of excitement. This illustrates the critical need for human oversight and domain knowledge when creating or applying dictionaries.